

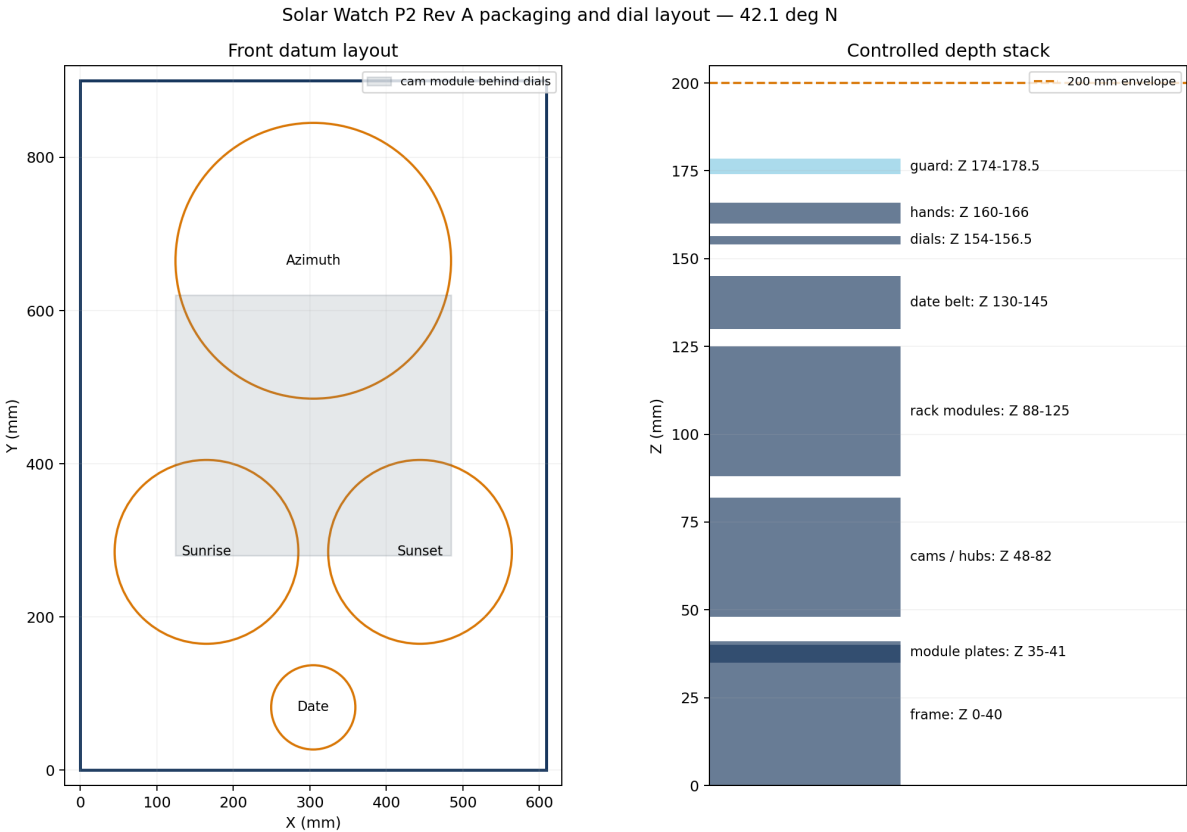
Solar Watch P2 Design Dossier

CNC-ready fixed-latitude astronomical display

REV	A	LATITUDE	42.1 deg N	RELEASE	FOR QUOTATION / PHYSICAL TEST PENDING
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Release statement

This release completes the digital P2 design and quotation package. Numerical results and kinematics are verified. Fabrication-dependent accuracy remains explicitly pending until the coupon, module, and integrated tests are completed.



P0 Numerical Reference and Validation

Status: Complete for P1

Reference year: 2025 (365 days)

Latitude: +42.1 deg

Longitude used for timestamps: 0 deg (UTC; it does not affect daylight duration)

Frozen convention

The reference event is the topocentric solar-center crossing of -0.833 deg at an ideal, unobstructed sea-level horizon. Azimuth is clockwise from true north. The reference year is deliberately non-leap so the P1 date cam has 365 equal daily increments. A future annual drive must address leap years; that is not part of P1.

The production calculation in `src/solar_watch/model.py` uses `pvlib 0.13.1`'s Python implementation of the NREL Solar Position Algorithm (SPA). SPA supplies the initial event estimate; a vectorized root refinement then locates the topocentric geometric-elevation crossing at exactly -0.833 deg. An automated test checks every sunrise and sunset to within 0.000001 deg of that altitude.

NREL describes SPA as a high-accuracy solar position algorithm with a reported solar-position uncertainty of about ± 0.0003 deg:

- [NREL SPA technical report](https://www.nrel.gov/docs/fy08osti/34302.pdf)
- [pvlib SPA implementation](https://pvlib-python.readthedocs.io/en/stable/reference/generated/pvlib.solarposition.spa_python.html)

Independent validation

`src/solar_watch/validation.py` independently computes every event in 2025 with `Skyfield 1.53` and the JPL DE421 planetary ephemeris. `Skyfield`'s documented sunrise definition is a solar center 0.8333 deg below the horizon; the script explicitly requests -0.833 deg to match this project:

- [Skyfield almanac documentation](https://rhodesmill.org/skyfield/almanac.html)

The committed JPL-derived table makes the 365-day comparison available offline. It can be refreshed with `solar-watch validate-ephemeris`; the first refresh downloads `de421.bsp` into `.cache/skyfield`, not into a system location.

Quantity	Maximum absolute difference	Requirement
Sunrise event time	0.002153 min (0.129 s)	0.1 min
Sunset event time	0.002144 min (0.129 s)	0.1 min
Daylight duration	0.000564 min (0.034 s)	derived diagnostic
Sunrise azimuth	0.000121 deg	0.1 deg
Sunset azimuth	0.000094 deg	0.1 deg

The model passes ACC-001 and ACC-002 for the P1 latitude and reference year. The full -60 . . +60 deg domain is boundary-tested but not claimed as a completed P3 validation sweep.

Outputs

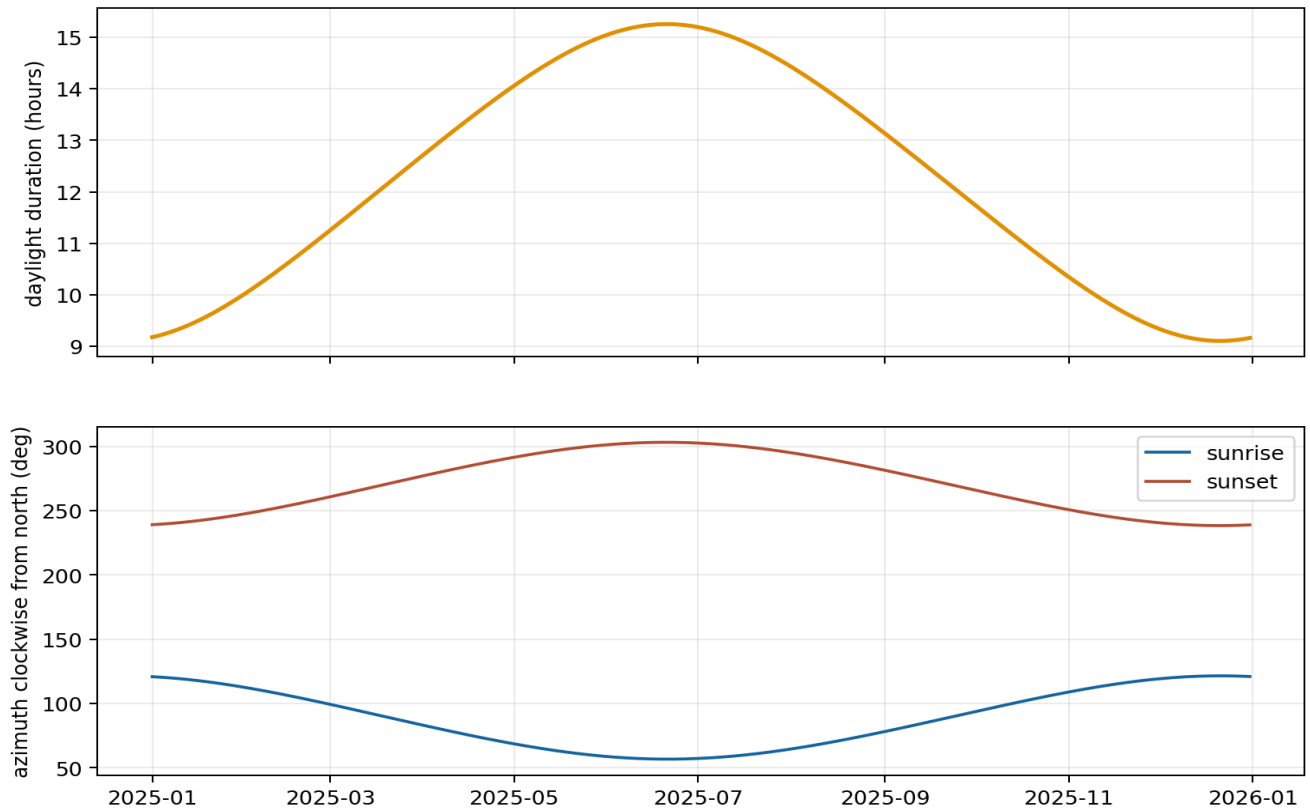
- `data/reference_42p1N_2025.csv`: the production full-year table.
- `data/skyfield_de421_42p1N_2025.csv`: independent full-year events.
- `data/validation_errors_42p1N_2025.csv`: row-by-row differences.
- `artifacts/plots/p0_reference_42p1N_2025.png`: annual functions.
- `artifacts/plots/p0_validation_errors_42p1N_2025.png`: error sweep.

The reference table retains timestamps and unrounded values. Cam generation uses those values directly; display rounding is never fed back into geometry.

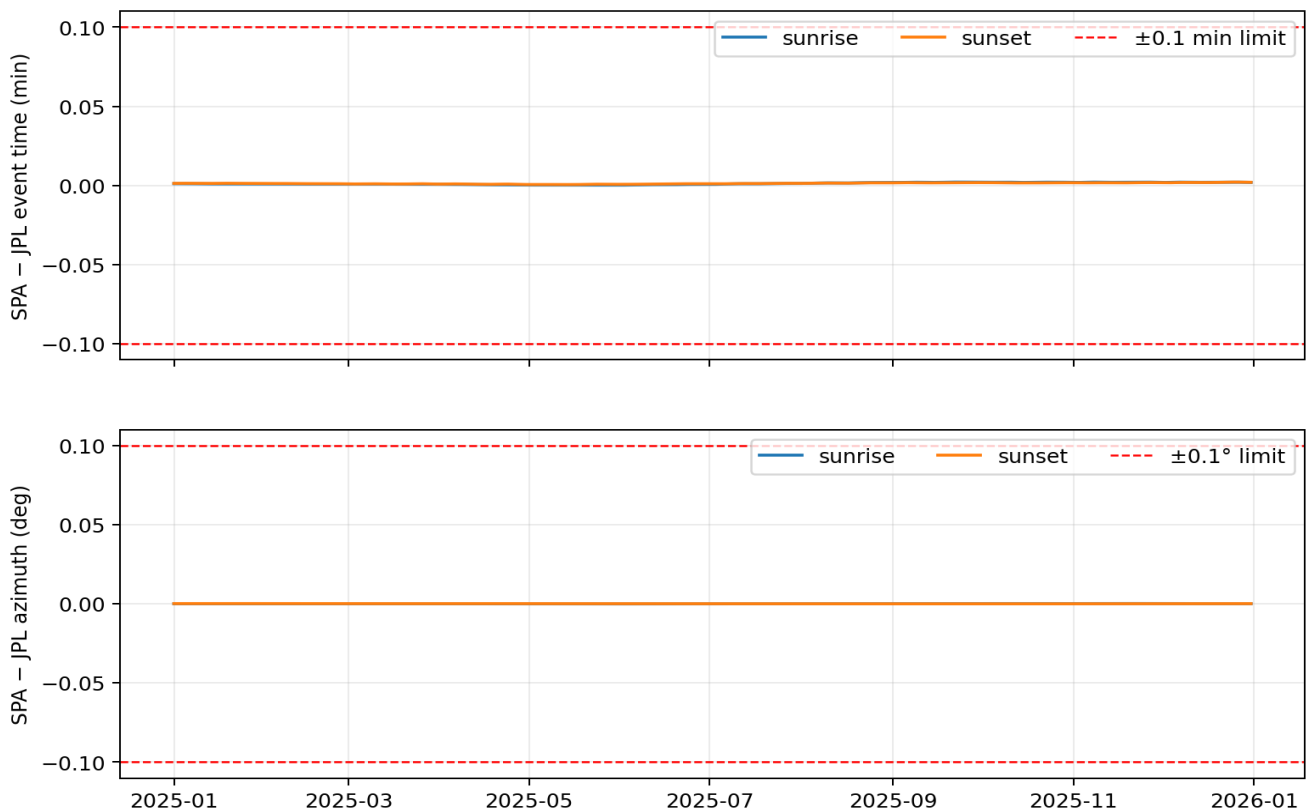
Symmetry finding

Using the event-specific solar positions, actual sunset azimuth differs from `360 deg - sunrise azimuth` by at most 0.277 deg in 2025. That is below the one-degree P1/P2 azimuth allocation, so mirrored sunset indication remains a valid later architecture choice. P1 nevertheless generates only the requested sunrise-azimuth cam and does not implement the P2 mirror mechanism.

Solar Watch P0 reference — 42.1° N, ideal horizon, 2025



Independent validation — NREL SPA vs Skyfield/JPL DE421



P2 Fixed-Latitude Design Basis

Status: Released for quotation, Rev A Latitude: 42.1 degrees north Units: millimetres unless noted

Scope and evidence boundary

P2 is a full-size, manually set, fixed-latitude astronomical display. One date input drives the P0-validated daylight-duration and sunrise-azimuth cams. The operator separately sets sunrise time. The mechanism displays sunrise time, mechanically calculates sunset time, and displays mirrored sunrise and sunset azimuths. The package intentionally stops before variable latitude, automatic civil-time correction, decorative dial artwork, or the P3/P4 mechanisms.

The solar model, cam profiles, linkage equations, nominal clearances, and CAD envelopes are digitally verified. Cutting accuracy, bearing fits, friction, preload, backlash, stiffness, and achieved indication error remain physical acceptance items. Decision 0006 records the deliberate P1 test bypass.

Display and frame

The controlled external envelope is 609 wide by 900 high by 200 deep. A 20-series metric T-slot perimeter supports three removable 6 mm 6061-T6 module plates and a 4.5 mm clear polycarbonate guard. The front layout uses these dial centers from the lower-left corner:

Display	Center X	Center Y	Diameter	Minor division
Sunrise azimuth / sunset azimuth	304.5	665	360	5 degrees
Sunrise time	165	285	240	5 minutes
Sunset time	444	285	240	5 minutes
Date setting	304.5	82	110	monthly reference

The five-minute time spacing is 5.236 mm on the dial rim. Five azimuth degrees occupies 15.708 mm. The functional scales are engraved or UV printed on 2.5 mm anodized aluminum; decorative artwork remains outside P2.

Cam stack and date datum

Both cams retain the P1 translating 16 mm roller geometry and validated scales:

Cam	Datum	Follower scale	Pitch radius	Surface radius range
Daylight duration	545 min	0.1308997 mm/min	48 mm nominal	generated manifest
Sunrise azimuth	55 degrees	0.6981317 mm/degree	48 mm nominal	generated manifest

Each 10 mm POM-C cam pilots on a 32 H7/h6 metal hub register, clamps with three M5 screws on a 40 mm bolt circle, and has a 4 mm indexing dowel at radius 20, 60 degrees counter-clockwise from the January 1 profile ray. The pair shares a 12 mm keyed date shaft. A split-clamp shaft collar gives continuous date phase adjustment of at least plus/minus 3 degrees after the dowel-defined nominal phase is established. Rotate the date control clockwise to advance the calendar while the followers remain on the positive-X datum ray.

The visible date control at (304.5, 82) drives the common cam shaft at (224.5, 375) through two 30-tooth HTD-5M pulleys and one 760 mm, 15 mm-wide belt. Equal tooth counts preserve 1:1 date phase. A smooth-back idler on a slot sets 10 to 20 N static belt tension and provides assembly tolerance. The date phase collar is downstream of the belt so belt indexing does not alter the cam datum. The belt is always approached clockwise during setting; bidirectional phase hysteresis is measured separately.

Cam faces are finish-machined together to 10.00 plus/minus 0.03 thickness. Profile tolerance is 0.05 relative to datum A (hub pilot); face parallelism and shaft-axis perpendicularity are 0.05. The follower surface is Ra 1.6 micrometre or better, with edges broken 0.2 maximum. Do not polish across the datum pilot.

Daylight-to-sunset mechanism

The sunrise knob drives a 15 mm pitch-radius pinion and the sunrise hand. Its rack travel is 94.248 mm for one 12-hour revolution. The P2 daylight cam directly encodes the required rack scale, so the daylight calibration lever is nominally 1:1; its annual rack range is approximately 48.31 mm.

Two opposed module-1 input racks constrain a freely rotating module-1, 15-tooth planet pinion carrier to their arithmetic mean. Carrier translation therefore equals one half of the summed rack positions. A third rack attached to the carrier drives a fixed-axis module-1, 15-tooth output pinion; its 7.5 mm pitch radius converts carrier translation back to the full angular sum. The sunset phase clamp removes the constant 545-minute cam datum. Nominal kinematics are:

```
x_sunrise = 15 rad(0.5 degrees/minute times sunrise_minutes)
x_daylight = 0.1308997 times (daylight_minutes - 545)
x_carrier = (x_sunrise + x_daylight) / 2
sunset_angle = x_carrier / 7.5 + phase
```

The racks are spring-biased against opposite tooth flanks by a 1 to 2 N extension spring. Rack guides use dry-running polymer bushings; oil is not permitted near POM cams. Carrier end stops provide 3 mm reserve beyond the calculated motion. The sunset zero collar provides plus/minus 15 degrees and the daylight lever slot provides plus/minus 2 percent span adjustment.

Azimuth mechanism

The P2 azimuth cam directly encodes 0.6981317 mm/degree. Its follower drives a rack through a nominal 1:1 adjustable calibration lever. The rack turns a module-1 pinion with 40 mm pitch radius (80 teeth), so one true azimuth degree produces one output degree. A preloaded 1:1 miter gear pair reverses the sunset output around the north-south meridian. The sunrise hand mounts on an 8 mm inner shaft and the sunset hand on a 14 mm OD by 8.1 mm ID outer tube. Two opposed spring washers remove axial clearance.

The sunrise zero collar allows plus/minus 5 degrees; the lever slot allows plus/minus 2 percent span. The sunset hand has an independent zero clamp so the two hands can be made symmetric at the equinox without disturbing the cam.

Both calibration levers use exact tangent-intersection geometry. A center-pivoted 180 mm lever has separate radial slots on each side. Input and output crosshead pins move on parallel guides at nominal offsets of minus and plus 75 mm, so their displacement ratio equals the ratio of guide offsets. Shifting the output guide from 73.5 to 76.5 mm provides 0.98:1 through 1.02:1 span without trigonometric scale error. Spring-biased slot rollers remove play; the calibrated guide is clamped by two bolts and jam nuts.

Followers, preload, and bearings

Both cams use sealed 16 mm OD by 5 mm bore track rollers, 7 mm nominal width, on shoulder screws. Each follower has 1 to 3 N positive preload throughout its travel and a mechanical retainer that prevents a disengaged roller from contacting the adjacent cam. All rotating display shafts run in replaceable flanged ball bearings or oil-impregnated bronze bushings located by machined module plates. Printed holes are not precision bearings.

Mechanical analysis and error allocation

The automated full-year sweep, pressure-angle loads, input torque, shaft/frame deflection, travel, interference clearances, backlash allocations, readability, and root-sum-square error budget are controlled by `p2-mechanical-analysis.md`. Estimated RSS is 0.768 minute and 0.557 degree against one-minute and one-degree targets. These are design estimates; physical module and integrated sweeps are required before the targets can be claimed.

Adjustment order

1. Inspect the hub pilot and shaft runout coupons.
2. Set date phase at January 1 with both cam dowels installed.
3. Set daylight and azimuth follower preload to 1 to 3 N.
4. Set daylight lever span using the minimum and maximum reference dates.
5. Set sunset zero at the 545-minute daylight datum.
6. Set azimuth lever span at the two solstice reference points.
7. Set sunrise azimuth zero, then set the reversed sunset hand for symmetry.
8. Lock all jam nuts, witness-mark clamps, and repeat the full sweep.

P2 Interface Control Document

Status: Released for quotation, Rev A All dimensions are millimetres. CAD is master for profile geometry; this file is master for functional interfaces and acceptance.

Coordinate system

Front view uses X right, Y up, and Z toward the viewer. The overall origin is the lower-left corner of the 609 by 900 envelope. Cam-fixed profile angle zero is the positive-X January 1 ray. The 4 mm indexing dowel is at radius 20 and 60 degrees counter-clockwise from that ray to avoid the three M5 clamp screws. Positive cam profile angles are counter-clockwise; the installed date knob advances clockwise.

Controlled interfaces

ID	Interface	Nominal	Fit / tolerance	Verification
I-01	Cam-to-hub pilot	32 diameter	H7 cam / h6 hub	bore gauge, micrometer
I-02	Cam clamp circle	3 x M5 on 40 PCD	true position 0.10 to I-01	CMM or pins
I-03	January 1 datum	4 reamed at radius 20, 60 deg CCW from profile zero	H7, position 0.05 to I-01	CMM or height gauge
I-04	Hub-to-date shaft	12 diameter keyed	H7 bore, commercial key	plug gauge
I-05	Cam face stack	10 each	plus/minus 0.03; parallel 0.05	micrometer
I-06	Roller follower	16 OD x 5 bore	commercial sealed bearing	certificate / measure
I-07	Time rack/pinion	module 1, 20 degree PA	quality 8 or better	supplier certificate
I-08	Sunrise pinion	30 tooth	15 pitch radius	roll test
I-09	Differential/output pinions	15 tooth	7.5 pitch radius	roll test
I-10	Azimuth pinion	80 tooth	40 pitch radius	roll test
I-11	Azimuth coaxial shafts	8 shaft, 14/8.1 tube	0.05 diametral running clearance	gauges
I-12	Module plate mounting	M6 clearance	6.6 diameter	plug gauge
I-13	Front guard	585 x 875 x 4.5	plus/minus 0.5 outline	tape / caliper
I-14	Overall envelope	609 x 900 x 200 maximum	must not exceed	assembled measurement
I-15	Date belt transmission	HTD-5M, 30T:30T, 760 length, 15 width	1:1; 10-20 N static tension	tension gauge / phase sweep

Datum scheme for custom parts

- Cams: datum A is the 32 mm pilot axis; B is the rear face; C is the January 1 dowel hole. The generated profile is basic relative to A and C.
- Hubs: A is the 12 mm shaft bore axis; B is the rear flange face; C is the hub dowel hole. Pilot runout is 0.025 total indicated relative to A.
- Module plates: A is the rear mounting face, B the lower long edge, and C the left short edge. Bearing and shaft locations are basic from B/C.
- Levers: A is the pivot bore; B is one broad face; C is the roller-slot center plane. Pivot-to-slot setting is adjustable and locked by two jam nuts.
- Shafts: A is the bearing journal axis; B is the front shoulder. Display hub runout is 0.025 total indicated relative to A.

General drawing notes

1. ISO 2768-mK applies unless specifically toleranced.
2. Deburr and break sharp edges 0.2 to 0.5; never roll or hand-blend cam profiles.
3. Aluminum is 6061-T6, clear anodized 8 to 12 micrometres after machining; mask bearing fits and electrical bonding points.
4. Cams are stress-relieved, machinable POM-C/acetel, black or natural. Machine both faces in one setup after roughing and allow 24 hours before finishing.
5. Shafts are 303 stainless, ground journals Ra 0.8 micrometre or better.
6. Polycarbonate guard edges are polished only after all holes are complete.

7. Supplier shall return dimensional inspection results for I-01 through I-05 and shaft runout. No material substitution without written approval.
8. STEP controls 3D form. DXF controls flat profiles. PDF drawing notes control tolerances, finishes, and inspection.

Assembly stack along Z

The frame rear plane is Z0. Module plates mount at Z35. The cam stack occupies Z48 through Z82 including hubs, spacers, and followers. Rack modules occupy Z88 through Z125. The date belt occupies Z130 through Z145. Dial faces lie at Z154 through Z156.5, hands at Z160 to Z166, and the guard inner face at Z174. The guard outer face at Z178.5 leaves at least 21.5 mm to the 200 mm maximum envelope for fasteners and stand-off variation.

Module placement in front coordinates

Module / axis	X	Y	Interface
Cam module plate center	304.5	450	P2-104, 360 x 340
Common cam/date shaft	224.5	375	local (-80,-75) on P2-104
Date control shaft	304.5	82	1:1 HTD link to cam shaft
Time module plate center	304.5	285	P2-108, 320 x 180
Sunrise shaft	165	285	local X -139.5 on P2-108
Sunset shaft	444	285	local X +139.5 on P2-108
Azimuth module / coaxial shaft	304.5	665	P2-109 center

The cam plate overlaps dial projections only in X/Y; its Z35-to-Z82 mechanism zone is behind the Z154 dial plane. Belt, spring, and fastener-head clearance is confirmed by the mandatory unguarded physical sweep.

Adjustment provisions

Adjustment	Range	Locking method	Witness mark
Date phase	plus/minus 3 degrees	split hub collar	shaft/hub line
Daylight span	plus/minus 2 percent	slotted lever, two jam nuts	lever scale
Sunset zero	plus/minus 15 degrees	split output collar	collar/shaft line
Azimuth span	plus/minus 2 percent	slotted lever, two jam nuts	lever scale
Sunrise azimuth zero	plus/minus 5 degrees	split pinion collar	collar/shaft line
Sunset azimuth zero	plus/minus 5 degrees	split hand hub	hub/tube line

P2 Mechanical Analysis and Error Budget

Status: Digitally verified Rev A; fabrication evidence pending

Full-year sweep

The committed analysis evaluates 365 dates at 48 sunrise settings in 15-minute steps: 17,520 states. The mathematical rack differential reproduces $(\text{sunrise} + \text{daylight}) \bmod 12$ hours with a maximum numerical kinematic error of $2.3\text{e-}13$ degree. The P0 mirror approximation differs from independently computed sunset azimuth by at most 0.277 degree. The full table is `data/p2_full_year_motion_sweep_42p1N_2025.csv`.

Travel	Minimum	Maximum	Allocated usable travel
Sunrise input rack	0.000 mm	92.284 mm at 11:45	100 mm
Daylight input rack	0.164 mm	48.471 mm	55 mm
Carrier output rack	0.082 mm	70.378 mm	78 mm
Azimuth rack	1.176 mm	46.451 mm	55 mm

Each guide receives at least 3 mm end reserve beyond the analyzed range. The sunrise rack additionally accommodates the untested 11:45-to-12:00 setting margin up to the 94.248 mm theoretical full turn.

Cam contact, force, and input torque

At the maximum specified 3 N follower preload, the daylight cam has 18.52-degree maximum pressure angle, 13.23 mm minimum pitch-curve curvature radius, 1.01 N maximum tangential follower force, and 65.14 N mm maximum ideal cam torque. The azimuth cam values are 17.58 degrees, 33.93 mm, 0.96 N, and 64.76 N mm. Both curvature radii exceed the 8 mm roller radius.

The two worst cam torques plus 40 N mm bearing/collar allowance total about 170 N mm. The date input is designed and acceptance-tested at 250 N mm; a 50 mm radius handwheel therefore requires no more than 5 N hand force and provides a 1.47 torque factor over the calculated nominal worst case. The sunrise input design torque is 75 N mm including 3 N working and 2 N preload forces, or 1.88 N at a 40 mm knob radius. Measure actual torque before installing the dials.

Deflection

An 8 mm stainless shaft with a 25 mm cantilever under 3 N deflects 0.00041 mm by Euler-Bernoulli beam theory. A 569 mm 20x40 extrusion span loaded at midspan with 10 N on its published weak-axis inertia deflects 0.046 mm. These estimates are below the 0.01 mm shaft and 0.10 mm frame allocations. Joint slip, plate flatness, and mounting compliance remain physical inspection items.

Clearance and interference

Automated planar and depth checks establish:

Check	Result	Requirement
Minimum dial-edge gap	39.0 mm	at least 20 mm
Guard inner face to hand stack	8.0 mm	at least 8 mm
Depth margin to 200 mm envelope	21.5 mm	at least 20 mm
Minimum cam profile-to-pilot radial wall	24.16 mm	at least 10 mm
Time / azimuth hand tip margin	15 / 15 mm	at least 10 mm

The assembly STEP and A01-A03 drawings additionally define module zones. The digital check cannot prove cable, spring, washer, or fastener-head clearance; the build procedure requires an unguarded slow sweep followed by an 8 mm guard clearance gauge check.

Backlash and hysteresis allocation

Commercial gear catalog backlash is not the achieved system lost motion. The design uses opposed flank springs on all three time racks and axial spring preload on the miter pair. The error budget assumes residual time-rack lost motion of 0.020 mm (0.306 minute at the 7.5 mm output radius), azimuth spur lost motion of 0.020 mm (0.029 degree), and miter lost motion of 0.030 mm at a 10 mm pitch radius (0.172 degree). These are acceptance limits, not vendor default claims. Directional tests must measure them after assembly.

Combined error budget

Independent terms are combined by root-sum-square for planning. This is not a guarantee because some fabrication errors can be correlated.

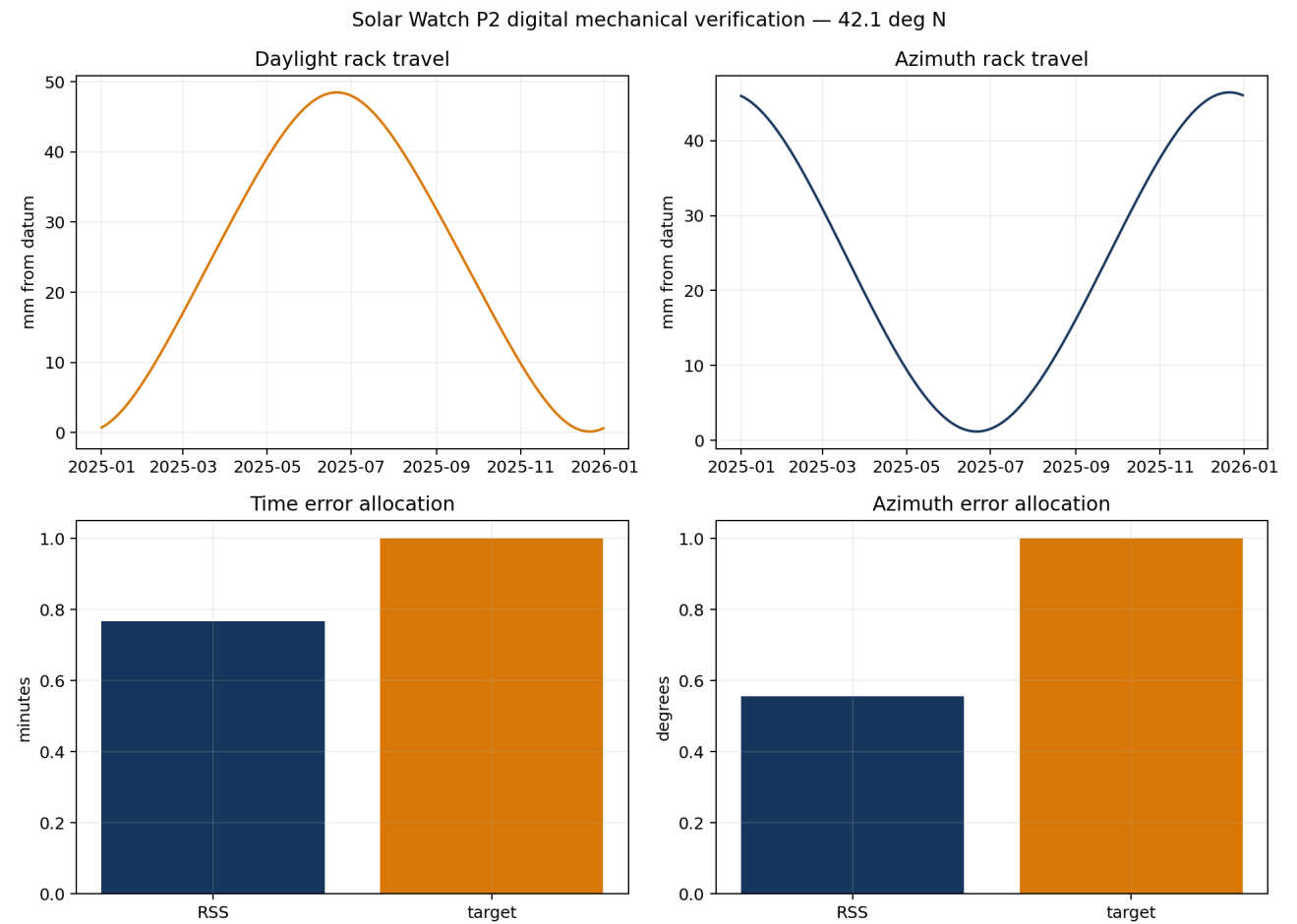
Contribution	Time, min	Azimuth, deg
P0 numerical reference	0.0022	0.00013
0.05 mm cam profile	0.382	0.072
0.025 mm hub runout	0.191	0.036

Contribution	Time, min	Azimuth, deg
0.020 mm follower/linkage	0.153	0.029
Preloaded spur/rack residual	0.306	0.029
Preloaded miter residual	not applicable	0.172
Calibration residual	0.200	0.150
Reading/parallax	0.500	0.500
Estimated RSS	0.768	0.557
Acceptance target	1.000	1.000

Functional time ticks are five minutes apart, 5.236 mm at the rim; a one-minute interpolation is approximately 1.047 mm. Five azimuth degrees span 15.708 mm; one degree spans 3.142 mm. Hands require a tip width at most 0.6 mm and a dial gap of 1.5 to 3 mm to keep the stated reading allocation credible.

Reproduction

Run `python -m solar_watch.cli analyze-p2 --root ..`. The command regenerates the full sweep, JSON summary, and mechanical verification plot. Automated tests assert kinematic exactness, travel coverage, clearance, deflection, RSS targets, and that physical validation remains marked pending.



P0/P1/P2 Verification Matrix

Requirement / deliverable	Evidence	Status
AST-001..004 conventions	model.py, conventions.md, P0 report	Pass by inspection/tests
AST-005 domain reporting	latitude and calendar boundary tests	Pass
AST-006 azimuth symmetry	365-day mirror-error column; max 0.277 deg	Pass for P1 allocation
ACC-001 event time ≤ 0.1 min	NREL SPA vs JPL DE421 full-year test; max 0.002153 min	Pass
ACC-002 azimuth ≤ 0.1 deg	same full-year test; max 0.000121 deg	Pass
ACC-003 one-minute mechanism	physical P1 measurements	Pending physical test; design target only
ACC-004 one-degree mechanism	physical P1 measurements	Pending physical test
ACC-005/006 repeatability/hysteresis	bidirectional print/test procedure	Pending physical test
MEC-003 segmentation	one-piece ~135 mm footprint; documented revision path	Pass for assumed 220 mm bed; printer confirmation pending
MEC-004 positive preload	1.3 N spring follower assumption and BOM	Pass by design; physical inspection pending
MEC-005 accessible adjustment	indicator zero, affine span, shaft phase adjustments	Pass by design; physical inspection pending
MEC-006/FAB-005 replaceable bearing/shaft	608 bearing and 8 mm metal shaft	Pass by design; physical inspection pending
FAB-001 consumer printing	watertight STL, ≤ 135 mm bounding boxes, no support	Digitally pass; physical print pending
FAB-003 identification	filenames, datum/ID hole patterns, revision, manifest	Pass
FAB-004 finish preserves datums	separate center bearing and internal datum hole; procedure	Pass by design
DOC-001/002 build traceability	build-record template	Pass for package; build record pending print
DOC-003 reproducibility	generator, pinned environment, offline verify script	Pass
DOC-004 calibration without CAD edits	coupon/zero/span/phase procedure	Pass by design
DOC-005 decision records	decisions 0001..0003	Pass
P2-001 shared manual date	keyed common shaft, cam hub/dowel interfaces	Pass by CAD; physical sweep pending
P2-002 manual sunrise input	30-tooth input pinion and direct sunrise hand	Pass by design; physical pending
P2-003 sunset arithmetic	full-year x 48 sunrise-setting software sweep	Pass digitally; physical pending
P2-004 mirrored azimuth	full-year linkage/mirror test, 1:1 reverser	Pass digitally; physical pending
P2-005 module calibration	three removable plates and module procedures	Pass by design inspection
P2-006 dial sizes	240 mm time and 360 mm azimuth DXFs	Pass digitally
P2-007 zero/span/phase	ICD adjustment table and range analysis	Pass by design; physical pending
P2-008 CNC definition	STEP/DXF/PDF, datum scheme, material/finish/inspection	Pass package inspection
P2-009 evidence boundary	design basis, procedure, matrix	Pass
P2-010 serviceability	guard-only access and independent plates	Pass by layout; physical review pending
P2 travel and kinematics	17,520-state full-year sweep	Pass digitally; max time arithmetic error $2.3e-13$ deg
P2 pressure/force/torque	generated cam-load and input-torque summary	Pass analysis; physical torque pending
P2 clearance/interference	dial, cam wall, hand, guard, and depth checks	Pass modeled checks; fastener/spring sweep pending
P2 deflection	shaft and weak-axis frame beam estimates	Pass allocation; assembled measurement pending
P2 error budget	numerical/cam/runout/linkage/backlash/calibration/readability RSS	Pass allocation at 0.768 min / 0.557 deg; physical pending

Software passing does not convert pending physical acceptance items into claims. P1 physical validation was deliberately bypassed by Decision 0006. The P2 package is ready for quotation, then coupon manufacture and module-first test.

Risk Register

Status: P2 Rev A Date: 2026-09-21

Scores use likelihood and consequence from 1 (low) through 5 (high). Priority is their product and is provisional until physical tests begin.

ID	Risk	L	C	Priority	Initial response
R-001	Cam and follower errors prevent one-minute indication.	3	5	15	CNC POM cam tolerance is 0.05 mm; run P2 pilot-fit and runout coupons before release, then report results without claiming the target prematurely.
R-002	Backlash causes different answers depending on setting direction.	3	4	12	Spring-preload rack flanks and followers; procedure records bidirectional hysteresis.
R-003	Variable-latitude surface cams bind or lose follower contact.	3	5	15	Analyze surface slopes; use roller followers; prototype the surface module independently.
R-004	One-minute accuracy cannot be read on the chosen dial.	4	3	12	Separate calculated accuracy from readable resolution; use large dials and five-minute markings.
R-005	Frame flexibility overwhelms cam accuracy.	2	4	8	20x40 perimeter/crossmembers and local 6 mm plates; measure deflection and shaft normality.
R-006	Rack differential adds excessive play or friction.	3	4	12	Use quality-8 module-1 parts, dual linear rails, opposed flank springs, and module-first acceptance.
R-007	Coaxial azimuth hands are difficult to fabricate and align.	3	3	9	Ground 8 mm inner shaft, 14/8.1 mm tube, replaceable bushings, wave-spring preload.
R-008	Astronomical convention differs from published sunrise data.	1	3	3	Mitigated for P1: convention frozen and full year passes NREL SPA vs JPL DE421 validation; retain ideal-horizon warning.
R-009	Atmospheric and terrain effects are mistaken for mechanism error.	3	3	9	State that the display represents a standardized ideal horizon.
R-010	POM cams move after rough machining or clamping.	3	4	12	Stress-relieved stock, rough/stabilize 24 h, finish pilot/profile together, coupon approval before production.
R-013	Incorrect bearing compensation or follower geometry corrupts the cam scale.	2	5	10	Source generates the inward normal offset for a 16 mm roller; automated scale, closure, pressure-angle, and mesh checks run on every verification.
R-014	Vendor silently smooths or resamples the cam profile.	2	5	10	STEP/DXF are geometry masters, CSV is inspection reference, RFQ forbids smoothing and requires 16-station results.
R-015	Bending or parallax at the large hands dominates reading error.	3	3	9	5052 hands, 1.5-3 mm dial gap, guard clearance, and integrated optical/angle check.

ID	Risk	L	C	Priority	Initial response
R-016	P1 test bypass leaves follower and preload behavior uncharacterized.	4	4	16	P2 coupon gate and separate cam/time/azimuth module tests are mandatory before integration.
R-011	Annual automatic drive requires more torque than a clock movement supplies.	4	3	12	Keep early date input manual; characterize torque before choosing a drive.
R-012	Uncontrolled scope shifts the project toward a wristwatch too early.	3	5	15	Treat the 36 by 24 inch demonstrator as the current product and wrist scale as a separate future project.

First risk-reduction experiments

1. Print a constant-slope calibration cam and measure follower repeatability.
2. Print one fixed-latitude annual azimuth cam.
3. Print one fixed-latitude annual daylight-duration cam at several output scales.
4. Compare roller and sliding followers.
5. Measure frame and shaft deflection under the intended preload.
6. Test a reversible output transmission and quantify lost motion.